Latency and Power Measurements of Neural Networks on FPGA-based RISC-V CPUs

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The ultimate goal of this project is to have a physical RISC-V system to evaluate power and latency behavior of different neural network implementations. Your task is to synthesize the CPU onto an FPGA board and enable it to execute simple tasks.

1 Resources

For the CPU you will use the Ibex [1] project, which implements a small $32\,\mathrm{bit}$ RISC-V core. There is a demo system [2] which uses a similar FPGA board. Feel free to research other generator projects that make use of this core. Communication between PC and CPU on the FPGA should happen via UART, an appropriate bootloader to load the program into the CPU can be chosen by you. The free license of Vivado should support our FPGA board [3], [4]. If you cannot use Vivado on your system you can get access to one of our lab PCs. Executable code for the Ibex CPU needs to be compiled with the appropriate compiler toolchain [5].

2 Expected Result

At the end of the project seminar you should be able to flash the FPGA with a bitstream that contains the RISC-V CPU and any required peripherals like, e.g., UART, memory. The system needs to be able to load a program and communicate with a PC via UART. You write a simple program that measures the delay of sending the sample via UART and the return of the inference result from the CPU and compare two provided network implementations.

References

[1] [Online]. Available: `https://github.com/lowRISC/ibex`

[2] [Online]. Available: `https://github.com/lowRISC/ibex-demo-system`

[3] [Online]. Available: `https://www.trenz-electronic.de/en/Nexys-A7-100T-FPGA-Trainer-Board-Recommended-for-ECE-Curriculum/29376`

[4] [Online]. Available: `https://svenssonjoel.github.io/pages/vivado-nexys-a7-getting-started/index.html`

[5] [Online]. Available: `https://github.com/riscv-collab/riscv-gnu-toolchain`